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20 August 1980
SUPERSEDING
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23 February 1978

MILITARY SPECIFICATION

BUCKLES; AND CLIPS, END, STRAP (FOR BELT, TROUSERS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers buckles and end clips (see 6.1).

1.2 Classification. Buckles and clips shall be of the following types, classes, sub-classes, and sizes as specified (see 6.2):

Type I - Buckle (closed front)
  Sizes:
    - 1-1/8 inch (29 mm)
    - 1-5/16 inch (33 mm)
    Class 1 - Yellow Brass
      Sub-class A - Lacquer Coated
      Sub-class B - Chromium Plated
    Class 2 - Nickel Silver

Type II - Buckle (open front)
  Size - 1-1/4 inch (32 mm)
    Class 1 - Yellow Brass
    Class 2 - Steel
  Size - 1-5/16 inch (33 mm)
    Class 3 - Red Brass

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type III - Clip
Sizes:
- 1 inch (25 mm)
- 1-1/4 inch (32 mm)
Class 1 - Yellow Brass
Sub-class A - Mirror Finish
Sub-class B - Black Enamel Finish
Class 2 - Nickel Silver
Size - 1-1/4 inch (32 mm)
Class 3 - Red Brass
Class 4 - Steel
Sub-class A - Brass Plated
Sub-class B - Black Enamel Finish

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-B-613	-	Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-C-320	-	Chromium Plating (Electrodeposited)
QQ-N-290	-	Nickel Plating (Electrodeposited)
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon
QQ-W-321	-	Wire, Copper Alloy
QQ-Z-325	-	Zinc Coating, Electrodeposited, Requirements For
UU-P-553	-	Paper, Wrapping, Tissue
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-676	-	Boxes, Setup
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

MILITARY

MIL-E-480	-	Enamel, Baking, Phenol-or Urea-Formaldehyde
MIL-F-495	-	Finish, Chemical, Black, For Copper Alloys
MIL-B-833	-	Belt, Trousers, Cotton Webbing, with Clip
MIL-B-17757	-	Boxes, Shipping, Fiberboard (Modular Sizes)

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STANDARDS

FEDERAL

FED-STD-151 - Metals; Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

DRAWINGS

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

2-4-57 - Buckles; and Clips, End, Strap (For Belt, Trousers); Type I, (Closed Front) Buckle, Sizes 1-1/8" and 1-5/16", Class 1 and 2
2-4-58 - Buckles; and Clips, End, Strap (For Belt, Trousers); Type III, Clip, End, Strap, Size 1" and 1-1/4", Class 1, 2, 3, and 4
2-4-63 - Buckles; and Clips, End, Strap (For Belt, Trousers); Die Cut Card for Mounting End Clip
2-4-64 - Buckles; and Clips, End, Strap (For Belt, Trousers); Type II, (Open Front) Buckle, 1-5/16 Inch Size, Class 3
4-1-436 - Buckles; and Clips, End, Strap (For Belt, Trousers); Type II, (Open Front) Buckle, 1-1/4 Inch Size, Class 1 and 2

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 122 - Copper-Nickel-Tin Alloy, Copper-Nickel-Zinc Alloy (Nickel Silver), and Copper-Nickel Alloy Plate, Sheet, Strip, and Rolled Bar
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.3, and 6.3).

3.2 Materials. The materials used shall conform to the applicable referenced specifications, standards, and requirements specified herein (see 6.6).

3.2.1 Copper base alloys.

3.2.1.1 Brass sheet or strip. Sheet or strip shall conform to copper alloy Nos. 230, 260 or 268 of QQ-B-613. Temper shall be as shown on drawings, or as specified herein.

3.2.1.2 Nickel silver. Nickel silver shall conform to UNS No. 75200 of ASTM B-122.

3.2.1.3 Wire, copper. Copper wire shall conform to alloy Nos. 260 or 270, 1/8 hard temper of QQ-W-321.

3.2.2 Steel, sheet and strip. Steel shall conform to cold rolled, chemical composition within the range of steel numbers 1009-1020 of QQ-S-698. Temper shall be as shown on drawings and as specified herein.

3.2.3 Lacquer. The lacquer shall be clear, synthetic and colorless, except as otherwise specified herein (see 3.4.3.4.1).

3.2.4 Enamel. Enamel shall be black conforming to type I or II of MIL-E-480.

3.3 Construction and performance. Buckles and clips shall be constructed in accordance with the design, details and dimensions shown on the applicable drawings and as specified herein. The buckles and clips shall be free from digs or gouges, burrs, slivers, sharp edges, dents or tool marks. The buckle clamps, rollers, and tongues shall be tested as specified in 4.4.3.2 and shall hold fast without belt slippage or deformation when a pull of 35 pounds (156 N) is applied.

3.3.1 Type I buckles.

3.3.1.1 Class 1. All components of the class 1 buckles except roller, shall be made from copper alloy Nos. 260 or 268 specified in 3.2.1.1, quarter-hard temper. The roller shall be made from copper wire specified in 3.2.1.3.

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3.3.1.2 Class 2. The front piece and roller of the class 2 buckle shall be made from nickel silver specified in 3.2.1.2. Wire for the roller shall be 1/8 hard temper. The back plate and clamp shall be made from copper alloy Nos. 260 or 268, quarter-hard temper brass specified in 3.2.1.1 or nickel silver specified in 3.2.1.2. When brass is used, the back plate and clamp shall be nickel plated conforming to class 1, grade G of QQ-N-290. Testing for nickel plating thickness shall be as specified in 4.4.3.2.

3.3.2 Type II buckles.

3.3.2.1 Size 1-1/4 inch (32 mm). The buckle shall be tested for function as specified in 4.4.3.2, with belt webbing as shown on Drawing 2-4-64. The webbing shall lie flat on top of tongue when tongue is snapped down in locked position.

3.3.2.1.1 Class 1. Class 1 buckles shall be made from copper alloy Nos. 260 or 268 specified in 3.2.1.1.

3.3.2.1.2 Class 2. Class 2 buckles shall be made from steel specified in 3.2.2.

3.3.2.2 Size, 1-5/16 inch (33 mm), class 3. Class 3 buckle components shall be made from copper alloy No. 230, quarter-hard temper brass specified in 3.2.1.1 except that quarter-hard temper is not required for the buckle body and ratchet back. The buckle shall be tested for function as specified in 4.4.3.2, with belt webbing as shown on Drawing 2-4-64. The webbing shall lie flat on top of tongue when tongue is snapped down in locked position.

3.3.3 Type III clips. Type III clips shall be formed with the smooth edge of the metal (see 6.4) on the outside of the clip. The size 1-1/4 inch (32 mm) clip when affixed to 1-1/4 inch (32 mm) wide belt webbing and tested as specified in 4.4.3.2 shall thread through a type I or II buckle, size 1-1/4 inch (32 mm) or 1-5/16 inch (33 mm) and of any class specified herein without binding or forcing. The size 1 inch (25 mm) clip when affixed to 1 inch wide belt webbing and tested as specified in 4.4.3.2 shall thread through a type I, 1-1/8 inch (29 mm) buckle of any class without binding or forcing.

3.3.3.1 Class 1. Class 1 clips shall be made from copper alloy No. 260 or 268, quarter-hard temper brass specified in 3.2.1.1.

3.3.3.2 Class 2. Class 2 clips shall be made from nickel silver specified in 3.2.1.2, and shall have a hardness of 75 to 78.5 on the Rockwell 15T scale. Testing for hardness shall be as specified in 4.4.3.2.

3.3.3.3 Class 3. Class 3 clips shall be made from copper alloy No. 230 quarter-hard temper brass specified in 3.2.1.1.

3.3.3.4 Class 4. Class 4 clips shall be made from steel specified in 3.2.2, 1/2 hard temper.

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3.4 Finish. The buckles and clips shall be finished as specified herein. Prior to finishing, all buckles and clips shall be thoroughly cleaned in accordance with commercial practice and shall be free from corrosion, foreign matter or discoloration.

3.4.1 Type I buckles.

3.4.1.1 Class 1 sub-class A. The outside face and two edge surfaces of the front piece and the exposed surface of the roller of class 1, sub-class A buckles shall be given a mirror-like polished finish. The back plate and clamp shall be bright-dipped, or ball-burnished finish. The complete buckle shall be lacquered with lacquer specified in 3.2.3, and as specified in 3.4.4.

3.4.1.2 Class 1, sub-class B. The outside face and two edge surfaces of the front piece and the exposed surface of the roller of class 1, sub-class B buckles shall be given a mirror-like polished finish. The back plate and clamp shall be bright-dipped, or ball-burnished finish. The buckle shall be nickel-plated in accordance with class 1, grade G of QQ-N-290 followed by chrome plating in accordance with class 1, type I of QQ-C-320 except the thickness shall be 0.000015 inch (0.00038 mm) minimum. The outside face and two edge surfaces of the front piece and the ends of the roller shall be polished to a uniform mirror-like finish.

3.4.1.3 Class 2. The outside face and two edge surfaces of the front piece and the exposed surfaces of the roller of class 2 buckles shall be given a satin finish. The scratch pattern shall be parallel to the long dimension edge of the buckle.

3.4.2 Type II, buckles.

3.4.2.1 Size 1-1/4 inch (32 mm).

3.4.2.1.1 Class 1. Class 1 buckles shall be given a black chemical finish conforming to MIL-F-495, except that the gloss shall be 40 maximmm. The gloss shall be estimated visually against a black panel that has a gloss of 40.

3.4.2.1.2 Class 2. Class 2 buckles shall be given a phosphate treated zinc plate conforming to type III, class 3 of QQ-Z-325, followed by bake enameling. The enamel shall be as specified in 3.2.4. The enamel coating shall level out to a uniform dry film thickness without orange peel, wrinkles, drops, streaks, peeling or flaking, specks, scratches, foreign matter or areas of no film.

3.4.2.2 Size 1-5/16 inch (33 mm).

3.4.2.2.1 Class 3. The tongue, ratchet back, and ratchet shall be bright-dipped or barrel-burnished. The face and edge surfaces of the body shall have a mirror-like polished finish or ball-burnished finish. The back of the body shall be buffed smooth.

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3.4.3 Type III clips.

3.4.3.1 Class 1.

3.4.3.1.1 Sub-class A (mirror finish). All outside surfaces shall be given a mirror-like polished finish and then lacquered with lacquer specified in 3.2.3 and as specified in 3.4.4. The inside surfaces of the clip may also be lacquered.

3.4.3.1.2 Sub-class B (black enamel finish). The entire clip shall be given a chemical dark oxide treatment in preparation for enameling. The final rinse stage shall be carefully conducted to remove alkaline residue and drying shall be thorough. The exterior surface and all edges shall be, and the interior may be, coated with enamel specified in 3.2.4 and baked. The finish shall be smooth, uniform and continuous, without orange peel, wrinkles, drops, streaks, peeling or flaking, specks, scratches or foreign matter. When tested as specified in 4.4.3.2, the enamel shall withstand clinching of the end clip to the belt without flaking, chipping or showing signs of cracks in the finish and without perceptible loss in adhesion to the point where the enamel can be removed by a thumbnail.

3.4.3.2 Class 2. All outside surfaces of the class 2 clip shall be given a satin finish and then lacquered with the lacquer specified in 3.2.3 and as specified in 3.4.4. The inside surface of the clip may also be lacquered. The scratch pattern shall be parallel to the short dimension edge of the clip.

3.4.3.3 Class 3. All outside surfaces of the class 3 clips shall be given a mirror-like polished finish.

3.4.3.4 Class 4.

3.4.3.4.1 Sub-class A (brass plated). All surfaces shall be completely covered with a commercial bright brass plating not less than 0.0002 inches (0.005 mm) thick when tested as specified in 4.4.2.1. The surfaces shall be coated with a colorless, baked epoxy lacquer.

3.4.3.4.2 Sub-class B (black enamel finish). All surfaces shall be given a phosphate treated zinc plate conforming to type III, class 3 of QQ-Z-325, followed by bake enameling. The enamel shall be as specified in 3.2.4. The enamel coating shall level out to a uniform dry film thickness without orange peel, wrinkles, drops, streaks or areas of no film. When tested as specified in 4.4.3.2, the baked on enamel shall adhere without flaking, chipping or showing signs of cracks in the finish.

3.4.4 Lacquering.

3.4.4.1 Type I, class 1 buckle; type III clips, classes 1, 2 and 4, sub-class A. The dry lacquer film shall be continuous, level, adherent, free from lint or other foreign inclusion, and not blushed, cloudy, hazy or have a rainbow effect. When tested as specified in 4.4.3.2, there shall be no change in the lacquered surface except for slight overall yellowing and there shall not be more than three (3) dark spots, each not exceeding 1/32-inch (1 mm) in the largest dimension.

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3.5 Marking for identification. All buckles shall be marked for identification with the letters "U.S." of the size as shown on the applicable drawings and manufacturer's identification of such known character as to be identifiable. The marking shall be permanently affixed by stamping or etching on the back surface of the buckle's back plate, clamp or ratchet as shown on applicable drawing. The identification marks shall be incorporated in such a manner that the surface opposite the markings shall show no signs of penetration or unevenness, or that distortion is not caused on any part of the buckle.

3.6 Workmanship. The buckles and clips shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the specified quality level.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects, specified in 4.4.3.1.1 and 4.4.3.1.2 and tested for the characteristics specified in table II. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, standards, and drawings unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection. Examination shall be made at any point or during any phase of the manufacturing process to determine whether the required operations are accomplished as specified. When applicable, in-process inspection shall be conducted to assure that the phosphate treated zinc plating has been accomplished and conforms to QQ-Z-325.

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4.4.2.1 Plating thickness test (class 4, sub-class A clip). Testing before lacquering shall be performed on brass-plated class 4, sub-class A clips to determine compliance with 3.4.3.4.1. The test method shall be a standard commercial thickness test. The lot shall be all unlacquered clips offered for inspection at one time. The sample unit shall be one clip. The inspection level shall be S-1 with an acceptable quality level (AQL) of 6.5 expressed in terms of defects per hundred units.

4.4.3 End item inspection.

4.4.3.1 End item examination. The lot shall consist of all completely finished buckles and clips as applicable, of one type, class, sub-class, size and finish offered for inspection at one time. The sample unit for this examination shall be one buckle or clip, as applicable.

4.4.3.1.1 Visual examination. Examination of the end item shall be made for the defects listed in table I. The inspection level shall be II with an AQL of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Color and finish	Not finished or the type of finish is not as specified	X	
	Discoloration clearly noticeable 1/	X	
	Any area of corrosion	X	
	Scratch pattern of satin-finished buckles and clips not in direction specified	X	
	Not lacquered (when required)	X	
	Black chemical, enamel, or lacquer finish is not adherent, e.g., blistered, peeling or flaking	X	
	Foreign matter imbedded in finish of surfaces exposed to view when in use 1/	X	
	Foreign matter imbedded in finish, but not exposed to view when in use		X
	Coating is not level, uniform, and continuous, e.g., runs, orange peel, wrinkles, drops or streaks 1/	X	
	Lacquer coating is blushed, cloudy, hazy, or has rainbow effect		X
	Finish is tacky to touch		X
	Buff, drag or mottled finish on surfaces exposed to view when in use 1/	X	
	Plating (when applicable) is broken, cut through, corroded, or erupted 1/	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Color and finish (cont'd)	Back, clamp, roller of buckle, or inside of buckle or clip is not bright and clean when required		X
	Penetration of markings to opposite surface, unevenness or distortion	X	
Design	Any characteristic not in accordance with the specified design	X	
Type, class and size	Not as specified or as shown on drawing	X	
Construction and workmanship	Component cracked, fractured, malformed, distorted, or bent	X	
	Dents, nicks, pores, abrasions, pits or scratches on surfaces exposed to view when in use 1/	X	
	Digs, gouges, burrs, slivers, sharp edges, or tool marks	X	
	Component missing	X	
	Operation omitted	X	
	Functioning component that is inoperative or will not operate as intended	X	
	Functioning component that requires abnormal force to operate but will function		X
	Edge not clean, e.g., sharp burr or fin	X	
	Smooth edge of clip not on outside	X	
	Prongs or teeth not sharply pointed where required	X	
	Any part misplaced, loose, or not in proper alignment	X	
Marking	Missing, illegible, or incomplete		X

1/ Noticeable at a normal inspection distance of 3 feet (1 m); if noticeable at less than 3 feet (1 m), classify as minor.

4.4.3.1.2 Dimensional examination. Examination shall be made of buckles and clips for compliance with dimensions specified on drawings. Any dimension that is not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

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4.4.3.2 End item testing. Testing of the completely finished buckles and clips shall be performed in accordance with table II for the characteristics specified. The sample unit shall be one buckle or clip.

TABLE II. End item tests

Characteristic	Requirement reference	Test method	Number determinations per sample unit	Results reported as pass or fail	Inspection level	AQL
Pull test (all types and classes of buckles)	3.3	4.5.3	1	X	S-1	6.5
Nickel plate thickness (type I, class 2 buckle)	3.3.1.2	QQ-N-290	1	X	S-1	6.5
Function (type II buckle)	3.3.2.1 and 3.3.2.2	4.5.2	1	X	S-1	6.5
Function (type III clips)	3.3.3	4.5.2	1	X	S-1	6.5
Hardness (type III, class 2 clip)	3.3.3.2	ASTM E 18	1	X	S-4	2.5
Adhesion of finish after clinching (type III, class 1 sub-class B clip)	3.4.3.1.2	4.5.5	1	X	S-1	6.5
Adherence of enamel (type III, class 4, sub-class B clip)	3.4.3.4.2	4.5.4	1	X	S-1	6.5
Lacquer resistance to liver of sulphur solution (type I, class 1 buckle; type III clips, classes 1, 2 and 4, sub-class A)	3.4.4.1	4.5.1	1	X	S-1	6.5

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4.4.4 Packaging inspection. An examination shall be made to determine that the preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with table III. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination of closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

TABLE III. Examination of packaging

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged or not as specified
Workmanship	Inadequate application of components such as: incomplete closure of container flap, loose strapping, improper taping, inadequate stapling Bulged, or distorted container
Content (exterior and interior)	Contents more or less than specified

4.4.5 Examination for palletization. An examination shall be made to determine that the palletization is in compliance with the requirements of section 5. Defects shall be scored as specified in table IV. The sample unit shall be one palletized unit load ready for shipment. The lot size shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 and the AQL shall be 6.5 defects per hundred units.

TABLE IV. Examination of palletization

Examination	Defect
Finished dimensions	Length, width, or height exceeding specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Lacquer test. The samples to be tested shall be immersed in a 2 percent by weight of chemically pure liver of sulphur (potassium sulphide) water solution, at a temperature of 100°F (38°C), for not less than 3 minutes. The samples shall then be removed and rinsed in warm 100° to 120°F (38° to 49°C), then cold 32° to 60°F (0° to 16°C), then hot 130° to 150°F (54° to 66°C) water, after which they shall be wiped gently with an absorbent cellulose material or whirled to remove residual moisture, and allowed to air dry at room temperature for a period of 1 hour. Any sample failing to meet the drying lacquer film requirements as specified in 3.4.4.1 shall be a defect.

4.5.2 Functional test of buckle, and clip (with clip attached to webbing). Type II buckles and type III end clips shall be tested by threading belt webbing (MIL-B-833) (see 6.5) with an end clip attached, through the buckle in position of normal use. The tongue after belt has been threaded through shall be pressed closed and remain closed until manually opened. Any sample failing to meet the webbing requirements as specified in 3.3.2.1 or 3.3.2.2, as applicable, or any sample failing to meet the threading requirements as specified in 3.3.3, shall be a defect.

4.5.3 Pull test. Buckles shall be tested for holding strength of clamp, rollers and tongue by inserting belt webbing (MIL-B-833) (see 6.5) in position of normal use as shown on applicable drawing. The webbing for the type I buckles shall be secured to the machine clamps at the roller end and clamp end. The webbing for the type II buckles shall be secured to the machine clamps at the tongue end and clamp end. Any sample failing to meet the pull requirements as specified in 3.3 shall be a defect.

4.5.4 Adhesion test. Class 4, sub-class A clips shall be tested for adhesion of enamel by clamping clips on belt webbing (MIL-B-833) (see 6.5). Any sample failing to meet adhesion requirements as specified in 3.4.3.4.1 shall be a defect. The same belt with class 4, sub-class A clip attached and used for test specified in 4.5.2 may be used for this test.

4.5.5 Clinching test, type III, class 1, sub-class B clip. The enameled end of the type III, class 1, sub-class B clip shall be clinched on 1-1/4 inch (32 mm) or 1 inch (25 mm) Government-furnished webbing so as to form a belt conforming to MIL-B-833 except that the length of the belt may be reduced to 8 inches (203 mm) for the purpose of this test. The width of the webbing shall depend upon the size of clip being tested. The clinching shall be done on a stamping press with the clearance between the flat platens at the bottoming point carefully adjusted to the maximum distance which will yield a smooth, closed end clip capable of passing through the assembled buckle without obstruction. The areas tested shall include the bend and adjacent surfaces. Any sample failing to meet adhesion requirements specified in 3.4.3.1.2 shall be a defect.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit pack.

5.1.1.1.1 Type I buckles. Four buckles of one size, and class only, shall be aligned side by side and then completely and compactly wrapped as a unit in a sheet of tissue paper approximately 15 inches (381 mm) in length and 10 inches (254 mm) in width conforming to type II, class 1 or 2 of UU-P-553. The wrap shall be accomplished in such a manner that the polished fronts of the buckles are protected by not less than three thicknesses of tissue. Alternatively, each buckle may be inclosed in a clear polyethylene film wrap having a thickness of 0.00125 (± 25 percent) inches (0.0318 mm).

5.1.1.1.2 Type II buckles. Each buckle shall be unit packed in an end opening coin type envelope made from 20 pound (33 g/m^2) minimum basis weight kraft paper (24 x 36-500). The size of the envelope shall be approximately 2-1/2 by 2-1/4 inches (64 by 57 mm). Alternatively, each buckle may be wrapped in polyethylene film as specified in 5.1.1.1.1.

5.1.1.1.3 Clips. Fifty clips of one size and class only, shall be mounted on a diecut, slotted paperboard card as shown on Drawing 2-4-63. The minimum thickness of the paperboard shall be 0.040 inch (1 mm).

5.1.1.2 Intermediate pack.

5.1.1.2.1 Buckles. Forty-eight type I buckles of one size and class only, or fifty type II buckles of one size and class only, unit packed as specified in 5.1.1.1, shall be packed in an intermediate set-up paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676, or a fiberboard box conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 175 of PPP-B-636. The inside dimensions of the box shall approximate 7 inches (178 mm) in length, 5-5/8 inches (143 mm) in width, and 2-1/2 inches (64 mm) in depth. Each paperboard box shall be secured with 2-inch (51 mm) minimum width gummed paper tape conforming to type III, grade B of PPP-T-45, applied girthwise around the center and overlapping not less than 2 inches (51 mm). Each fiberboard box shall be closed in accordance with method II of the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.1.1.2.2 Clips. Six-hundred clips of one size and class only, unit packed as specified in 5.1.1.1, shall be packed in an intermediate set-up paperboard box or fiberboard box constructed and closed as specified in 5.1.1.2.1. The inside dimensions of the box shall approximate 8-7/8 inches (225 mm) in length, 8-5/8 inches (219 mm) in width, and 6-3/4 inches (172 mm) in depth. Approximate dimensions are furnished as a guide only.

5.1.2 Commercial. Buckles and clips shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

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5.2.1 Level A packing.

5.2.1.1 Buckles. Five hundred and seventy-six type I buckles of one size and class only, or six-hundred type II buckles of one size and class only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A intermediate packs shall be packed on end, six in length, two in width, and one in depth within a shipping container. Inside dimensions of each shipping container shall approximate 15-1/2 inches (394 mm) in length, 11-5/8 inches (295 mm) in width, and 7-1/8 inches (181 mm) in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.1.2 Clips. Seven thousand two-hundred clips of one size and class only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A intermediate packs shall be packed flat, two in length, two in width, and three in depth within a shipping container. Inside dimensions of each shipping container shall approximate 18-1/4 inches (464 mm) in length, 18 inches (457 mm) in width, and 20-1/4 inches (514 mm) in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636.

5.2.2 Level B packing.

5.2.2.1 Buckles. Five hundred and seventy-six type I buckles of one size and class only, or six hundred type II buckles of one size and class only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed on end, six in length, two in width, and one in depth within a shipping container. Inside dimensions of each shipping container shall approximate 15-1/2 inches (394 mm) in length, 11-5/8 inches (295 mm) in width, and 7-1/8 inches (181 mm) in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.2.2 Clips. Seven thousand two-hundred clips of one size and class only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A intermediate packs shall be packed flat, two in length, two in width, and three in depth within a shipping container. Inside dimensions of each shipping container shall approximate 18-1/4 inches (464 mm) in length, 18 inches (457 mm) in width, and 20-1/4 inches (514 mm) in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

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5.2.2.3 Weather-resistant fiberboard containers. When specified (see 6.2), shipping containers packed with buckles or clips shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.4.

5.2.3 Commercial packing. Buckles and clips, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Palletization. When specified (see 6.2), buckles and clips, packed as specified, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be affected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern to be used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract, intermediate boxes, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The buckles and clips are intended for use on web trouser belts.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type, class, sub-class, and size required (see 1.2).
(c) When a first article is required (see 3.1, 4.3 and 6.3).
(d) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing of buckles or clips (see 5.2.2.3).
(f) When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article shall be a preproduction sample and should consist of six sample buckles or twelve clips, as applicable. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Punching of type III clips. Punching of the blank for the clip produces a typical sheared edge which is smooth on one side and rough or sharp on the opposite side (see 3.3.3).

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6.5 Samples. For access to web belts for tests, address the procuring activity issuing the bids.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.2).

6.7 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 11

Review activities:

Army - MD
Navy - MC
Air Force - 99
DLA - CT

Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-B-1963H
BUCKLES; AND CLIPS, END, STRAP (FOR BELT, TROUSERS)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☒ VENDOR ☐ USER ☐ MANUFACTURER

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